

Wave Motion

Open organ pipe

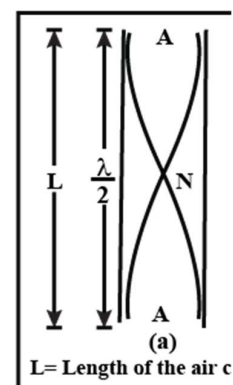
The pipe in which the both of its ends are open is called open organ pipe. Flutes is the example of organ pipe.

1. First mode of vibration/First harmonic/Fundamental frequency

In the mode of vibration in the organ pipe, two antinodes are formed at two open ends and one node is formed in between them. If length of the pipe be 'l' and be the wavelength of wave emitted in this mode of vibration. Then,

$$l = \frac{2\lambda_0}{4} \Rightarrow \lambda_0 = 2l$$

Frequency $f_s = \frac{v_s}{2l}$



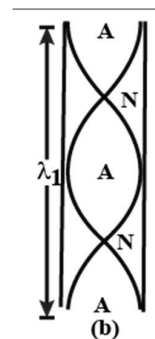
2. Second mode of vibration/First overtone /2nd harmonics

In the second mode of vibration in the open organ pipe, there antinodes are formed at two ends and two nodes between them.

If 'l' be the length of the pipe and λ_1 be the wavelength of wave emitted in this mode of vibration. Then,

$$l = \lambda_1$$

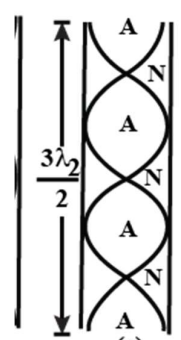
Frequency $f_1 = \frac{v_s}{l} = 2f_0$



3. Second mode of vibration/second overtone/3rd Harmonics

$$l = 3/2\lambda_2$$

Frequency $f_2 = \frac{3v_s}{2l} = 3f_0$



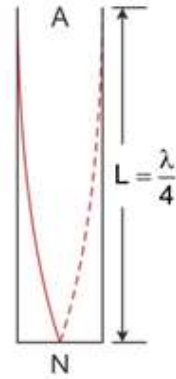
Closed organ pipe

Closed organ pipe is the one in which only one end is open and the other is closed and then sound is passed.

1. Fundamental frequency

$$l = \frac{\lambda_0}{4} \Rightarrow \lambda_0 = 4l$$

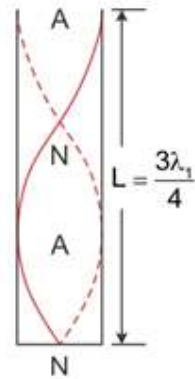
Frequency $f_0 = \frac{v_s}{4l}$ -First harmonics



2. First overtone

$$l = \frac{3\lambda_1}{4} \Rightarrow \lambda_1 = \frac{4l}{3}$$

$f_1 = \frac{3v_s}{4l} = 3f_0$ -Second harmonics



3. Second overtone

$$l = \frac{5\lambda_2}{4} \Rightarrow \lambda_2 = \frac{4l}{5}$$

$f_2 = \frac{5v_s}{4l} = 5f_0$ -Third harmonics

